

Grounding biological data to databases

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Abstract

Single-cell RNA-seq produces cell type marker genes, yet they often fail to reproduce across studies. This failure is, in part, a technical artifact of how markers are reported. Flat cell type–gene pairs discard the comparisons that produced them. In **LLMarkers**, a benchmark we curated from seven studies, reported markers could not be fully reproduced from the underlying differentially expressed genes. A large language model (LLM) recovered them roughly fourfold more accurately by extracting them from manuscript text than by generating or selecting them. We therefore built **mrkr**, a tool that uses an LLM to extract reported markers from papers and map their labels deterministically to stable identifiers, and applied it to 979 papers. Stable identifiers connected markers that label matching missed. But stable identifiers are insufficient for reusable markers. Changing only the comparison group of cells reduced recovery of reported markers from 74% to 21%. These results suggest a definition for a marker gene that is valid relative to the comparisons that produced it. Reusable markers require recording the comparison group of cells with stable identifiers, which our `.onto` format provides.

1 Introduction

Single-cell RNA sequencing identifies cell types by comparing gene expression across cells in an experiment [Trapnell, 2015, Wagner et al., 2016, Regev et al., 2017, Heumos et al., 2023]. A marker gene is therefore not an inherent property of a cell type. It is the result of a comparison in which a gene ranks as a marker only relative to gene expression measured in the other cells in the experiment [Pullin and McCarthy, 2024]. Adding groups of cells to a single-cell analysis can change which genes are markers. Yet markers are routinely treated as atomic and reusable, reported in papers and curated in databases as flat cell type–gene pairs, as if the comparison that produced them were secondary to the result [Franzén et al., 2019, Hu et al., 2022].

This reporting practice creates two problems. The first is that labels are unstable [Miller et al., 2020, Osumi-Sutherland et al., 2021]. The same population of cells may be called “plasma B cell” in one paper and “plasma cell” in another, and “red blood cell” in one paper and “erythrocyte” in another. Genes also appear under different names, such as “DNGR-1” and *CLEC9A*. Stable identifiers can resolve these differences—the Cell Ontology for cell types and Ensembl for genes [Osumi-Sutherland et al., 2021, Cunningham et al., 2021]—but papers rarely report them [Lubiana et al., 2022].

The second problem is that stable identifiers do not state the comparison. Knowing that two studies both

report *CD8A* (ENSG00000153563) as a marker for CD8-positive, alpha-beta T cells (CL:0000625) does not tell us which cells each study compared. Since markers are defined by comparison, a marker stripped of its comparison no longer states what it was shown to distinguish. A marker derived from sorted peripheral blood mononuclear cells and one derived from a whole-tissue atlas may share a name but arise from different comparisons.

These two reporting problems are distinct from, but related to, biological and technical variability in scRNA-seq data. Assay chemistry, sequencing depth, quantification, normalization, replicate structure, donor variation, disease state, and differential-expression methods can change the status of a marker [Heumos et al., 2023, Soneson and Robinson, 2018, Squair et al., 2021, Krzak et al., 2026]. The flat cell type–gene reporting scheme records neither these experimental details nor the comparison group of cells.

Effective marker reuse therefore requires stable identifiers to establish that studies refer to the same cell type and gene, and the comparison group of cells to establish what each reported marker gene was shown to distinguish.

In this work, we ask how well reported marker genes can be recovered, connected, and reused across studies, and what information must be retained to support reuse. We proceed in three steps. First, using a manually curated benchmark, we compare reported markers with their differential-expression strength in the underlying data and test whether a large language model (LLM) can generate, select, or extract the same cell type–gene associations. Second, we introduce **mrkr**, a tool that extracts reported cell type–marker gene associations from papers,

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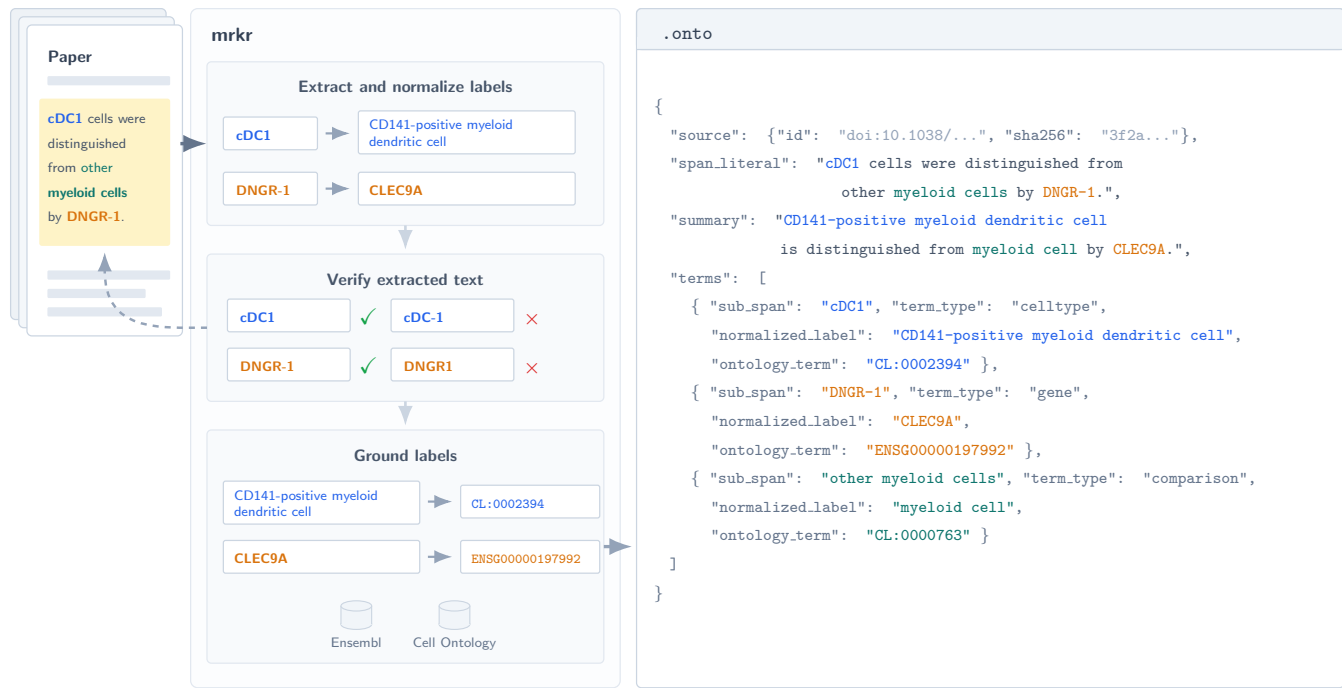


Figure 1: mrkr converts reported markers into machine-readable .onto records. The LLM extracts the supporting sentence and the reported cell type and gene labels, then proposes normalized labels. Exact-string validation accepts cDC1 and DNNGR-1 and rejects cDC-1 and DNNGR1, which do not occur in the paper. Deterministic lookup maps normalized labels to Ensembl and Cell Ontology identifiers; the LLM does not assign identifiers. The .onto record retains the supporting sentence, reported and normalized labels, and identifiers. Blue denotes the cell type, orange the marker gene, and teal the comparison group of cells.

normalizes each reported label, and maps it deterministically to stable identifiers from the Cell Ontology and Ensembl (**Figure 1**). Each association is stored in the .onto format, which records the reported and normalized labels, stable identifiers, supporting sentence, and any comparison group of cells named in the paper. Third, we apply mrkr to 979 papers. Stable identifiers connect reported associations that exact label matching misses, but the sets of reported markers still diverge across studies. A controlled analysis that holds the target cells fixed while changing the comparison groups shows why: identifiers establish which genes and cell types can be connected, while the comparison group establishes what each marker was shown to distinguish. Effective marker reuse requires both.

2 Results

Differential-expression rankings only partially reproduce reported markers

If scRNA-seq marker genes are fully derived from data, then reported markers should be recoverable from the differential expression computed in each study. Testing this requires a record of which markers a paper reported. We manually curated 2,528 reported cell type–gene associations from seven scRNA-seq studies spanning six tissues, together with each study’s differential-expression

(DE) tables, and call this benchmark **LLMarkers**. The benchmark covers 168 cell types, 641 genes, and 1,560 unique cell type–gene associations (**Supplementary Table S1**).² Most markers were reported only in figures: 1,151 of the 1,560 associations (74%), with 256 (16%) only in text and 153 (10%) in both (**Figure 2A**).³

Reported markers could not be fully reconstructed from the DE tables. The fraction of reported markers present in a study’s DE tables ranged from 18% in the embryonic skeletal development study [He et al., 2021] to 97% in the idiopathic pulmonary fibrosis study [Adams et al., 2020] (**Figure 2A**). In the Emont adipose study, the authors report *PDGFRA* as a marker of subpopulations hASPC1 through hASPC6, but the DE table for those subpopulations includes it only for hASPC1.⁶ The same marker could also appear in two tables with fold changes in opposite directions. Among 544 marker gene associations found in at least two DE tables from the same study, 46 (8.5%) were upregulated in one table and downregulated in the other.⁷ The direction of a reported marker depends on which cells it was compared to, and each table is a different comparison.

Markers present in the DE tables were not simply the top-ranked genes. Among markers reported in text, 53% appeared in the top 50 differentially expressed genes (DEGs); among markers reported in figures, 50%. The top 100 DEGs contained 66% and 62%.⁸ No single

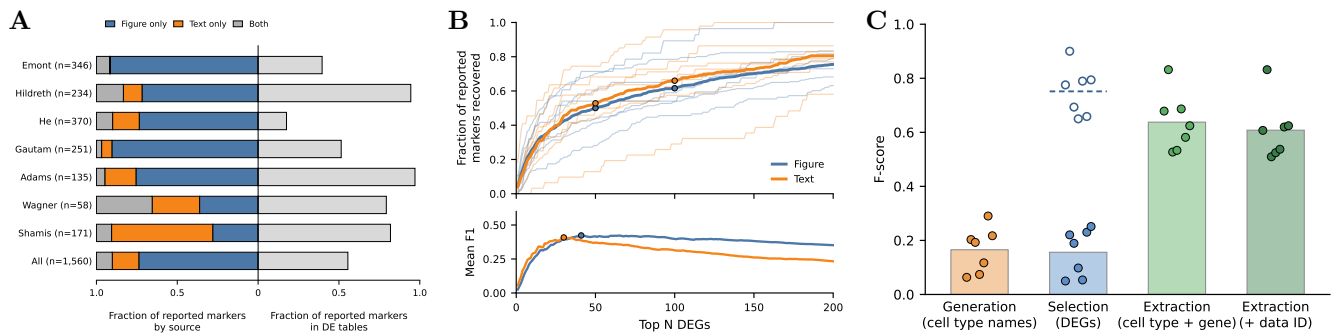


Figure 2: Reported markers are not fully recovered by DEG rank but can be extracted from papers. **A.** For each study in the LLMarkers benchmark, the fraction of reported markers by source (left) and the fraction of reported markers present in that study's DE tables (right). **B.** Fraction of reported markers recovered among the top N DEGs. Thin lines are studies, split by source; bold lines pool all studies; dots mark $N = 50$ and $N = 100$. The lower panel shows the mean F-score across studies at each cutoff, with each maximum marked. **C.** F-score for LLM curation tasks: marker generation from cell type names, selection from ranked DEGs, and extraction from manuscript text. Bars are means across studies and points are studies. Open circles show the F1 selection would reach if the model chose perfectly from each study's DEG list; the dashed line is their mean. All three tasks were scored against the markers reported in text (Methods).

rank cutoff separated reported markers from the rest. Averaged across studies, F-score peaked at only 0.41 for markers reported in text and 0.42 for markers reported in figures, near cutoffs of 30 and 40 genes (**Figure 2B**), and per-study optimal cutoffs ranged from 21 to 186 genes.⁹ Within the Hildreth study alone, rank and precision–recall distributions varied by cell type (**Supplementary Figure S1**). Differential expression is often described as a marker selection process, but our results indicate that authors do not simply select the top-ranked genes, so marker selection cannot be reconstructed from ranking the underlying data alone.

LLMs extract markers better than they generate or select them

Marker genes could not be fully reconstructed from DEG rankings, so we asked whether an LLM could produce them. We compared three LLM tasks. The first prompted the model to generate markers from cell type names alone. The second prompted the model to select markers from ranked DE tables. The third prompted the model to extract reported markers from manuscript text (**Figure 2; Supplementary Table S2**). The tasks differ in how much of each paper the model sees (none for the first task, the DE tables for the second, and the full text for the third).

Extraction recovered reported markers more accurately than generation or selection. Generation from cell type names reached a mean cell type–gene F1 of 0.17 against the markers reported in text.¹⁰ Selection from ranked DE tables reached 0.16 at the best input depth for each study.¹¹ Extraction reached a mean F1 of 0.64, and 0.61 when the model also identified the DE table supporting each association.¹² Extraction is scored against the same text it reads, so part of its advantage is by

construction. That construction is the goal of curation: to preserve what the authors reported rather than predict it. As a consequence, accurate marker gene recovery required reading the paper, which supports using LLMs to extract and normalize reported markers.

We implemented LLM-based marker extraction in **mrkr**, a command-line tool that converts manuscript text into a validated **.onto** file (**Figure 1**). Each retained association links the supporting sentence to one organism, one cell type label, at least one marker gene, the marker direction, and any tissue or comparison group of cells named in the paper. Validation checks that the supporting sentence and every extracted term occur exactly in the manuscript (Methods). The LLM extracts named cell types and genes and produces normalized labels that are mapped, deterministically, to stable identifiers from Ensembl, NCBI Taxonomy, Cell Ontology, and UBERON.¹³ This separation isolates the variability of author naming from the assignment of stable identifiers, so label mappings can be audited and revised without changing what the paper reported.

Stable identifiers connect markers across papers

To test whether stable identifiers connect markers across papers, we applied **mrkr** to 504 bioRxiv preprints and 475 papers linked to the Human Cell Atlas. Of the 979 papers, 721 reported at least one cell type–gene association, totaling 11,336 associations of 27,713 marker genes. 586 of the 11,336 associations (5.2%) named a comparison group of cells.¹⁴

Of the 27,713 reported marker genes, 26,919 (97%) mapped to an Ensembl identifier.¹⁵ Cell type labels mapped less often. We accepted a Cell Ontology identifier only when a normalized label matched a canonical label

or an official exact synonym. Of 3,355 distinct cell type labels, 294 matched a canonical label, 112 matched an exact synonym, and the remaining 2,949 (88%) matched neither. The 406 accepted labels (12%) covered 363 identifiers.¹⁶ Gene names largely followed a shared nomenclature while cell type names did not. But despite the low mapping rate, stable identifiers showed how variable naming conventions collapse onto single cell types (**Figure 3A**). “T cell,” “T-cell,” “T lymphocyte,” and “T-lymphocyte” all mapped to CL:0000084. Overall, 324 of the 363 identifiers received one reported label, 37 received two, and two identifiers received four.¹⁷

To compare markers across papers, we first pooled the genes reported for each normalized cell type label in each paper. We call this set the paper’s marker gene panel. Two panels from different articles that used the same normalized label form a panel pair. Keeping panels with at least three genes produced 3,454 panels and 10,437 panel pairs.¹⁸

Stable identifiers connected panels that exact label matching missed (**Figure 3B**). Twelve identifiers connected 248 panel pairs with different normalized labels, such as “plasma B cell” and “plasma cell,” and 113 of the 248 (45.6%) shared at least one marker. Panel pairs whose labels matched exactly shared a marker at a similar rate (49.2% with a shared identifier, 54.1% without, after matching by paper source and panel size). Panel pairs with different identifiers shared a marker 1.6% of the time.¹⁹ Marker sharing never exceeded 55% in any group because papers report different cells in different tissues and disease states, under different cell type comparisons. Therefore, stable identifiers recover associations that exact labels miss but a shared identifier does not establish that the sampled cell groups are biologically identical or that the cell type and markers associated with it are reusable.

Marker panels vary with cellular context

The cellular context of a marker includes the tissue, the disease state, and the cell types observed in an experiment, among other technical and biological covariates. The comparison group of cells is one measurable component of this context. If reported markers depend on the cellular context, then papers that observed similar cell types should report more similar markers. Cell type labels reported in a paper provide a limited proxy for the cell types each paper observed (**Figure 3C**). Across the 10,437 panel pairs, marker sharing was weakly positively associated with the similarity of the co-reported labels (Pearson $r = 0.149$).²⁰ Papers that reported more similar sets of cell types tended to report more similar marker panels. The association is weak in part because co-reported labels are an imperfect proxy in that they do not fully identify the cellular context that generated the marker.

To test whether persistent markers emerge as curated marker sets accumulate, we took the 37 normalized la-

bels reported in at least ten papers, added panels one paper at a time in random order, and asked whether any marker remained in every panel. Each additional paper contributes markers from a different cellular context. If a cell type label named one fixed marker panel, combining papers would preserve a shared core of markers. Shared markers were lost as papers accumulated (**Figure 3D**). After two papers, an estimated 51.8% of labels retained at least one shared marker. The fraction fell to 28.8% after three papers, 11.3% after five, and 1.7% after ten. Repeating the analysis with 27 accepted Cell Ontology identifiers gave nearly identical results.²¹ A cell type label or identifier does not specify a marker panel that remains stable as papers accumulate.

Changing the comparison groups changes marker recovery

The marker accumulation results point to the necessity of cellular context for defining a cell type marker. But isolating the effect of cellular context requires holding the cell types and the measurements fixed while changing only the comparison group of cells. This is generally challenging but the Cell Annotation Platform (CAP) of the Human Cell Atlas provides a close enough proxy to this design. They host data for one study that contains a liver myeloid analysis nested within an all-cell liver analysis of the same cells. The myeloid analysis separates ten myeloid cell types, while the all-cell analysis, on the same cells, adds non-myeloid cells and groups the myeloid cells under coarser labels, resulting in 17 total cell types. Comparing the two datasets therefore changes the groups entering each differential-expression calculation while the measurement stays fixed.

In the CAP myeloid cell types, reported markers were recovered more often in the analysis they were curated from (**Figure 4A**). Across the ten myeloid labels, one-vs-rest differential expression within the myeloid analysis recovered a mean of 74% of reported markers in the top 100 genes. The all-cell analysis recovered 21%. Recovery was higher in the myeloid analysis for every label.²⁴ The two analyses also produced different rankings: for every label, the top 100 DEGs of the two analyses overlapped by less than half (Jaccard 0.05–0.48; **Figure 4A**).²⁵ Defining each all-cell target group by the exact barcodes of its myeloid cells, rather than by the study’s all-cell labels, gave the same result: a mean recovery of 22% (**Figure 4B**).²⁶ The drop is therefore driven by the change in comparison groups, not by differences in how the cells were labeled. Changing the comparison groups changed which reported markers were recovered, while the rest of the cellular context stayed fixed.

Marker sets are defined by their comparison groups

The dependence of markers on comparison groups can be stated exactly. For a study s , partition p , target cell

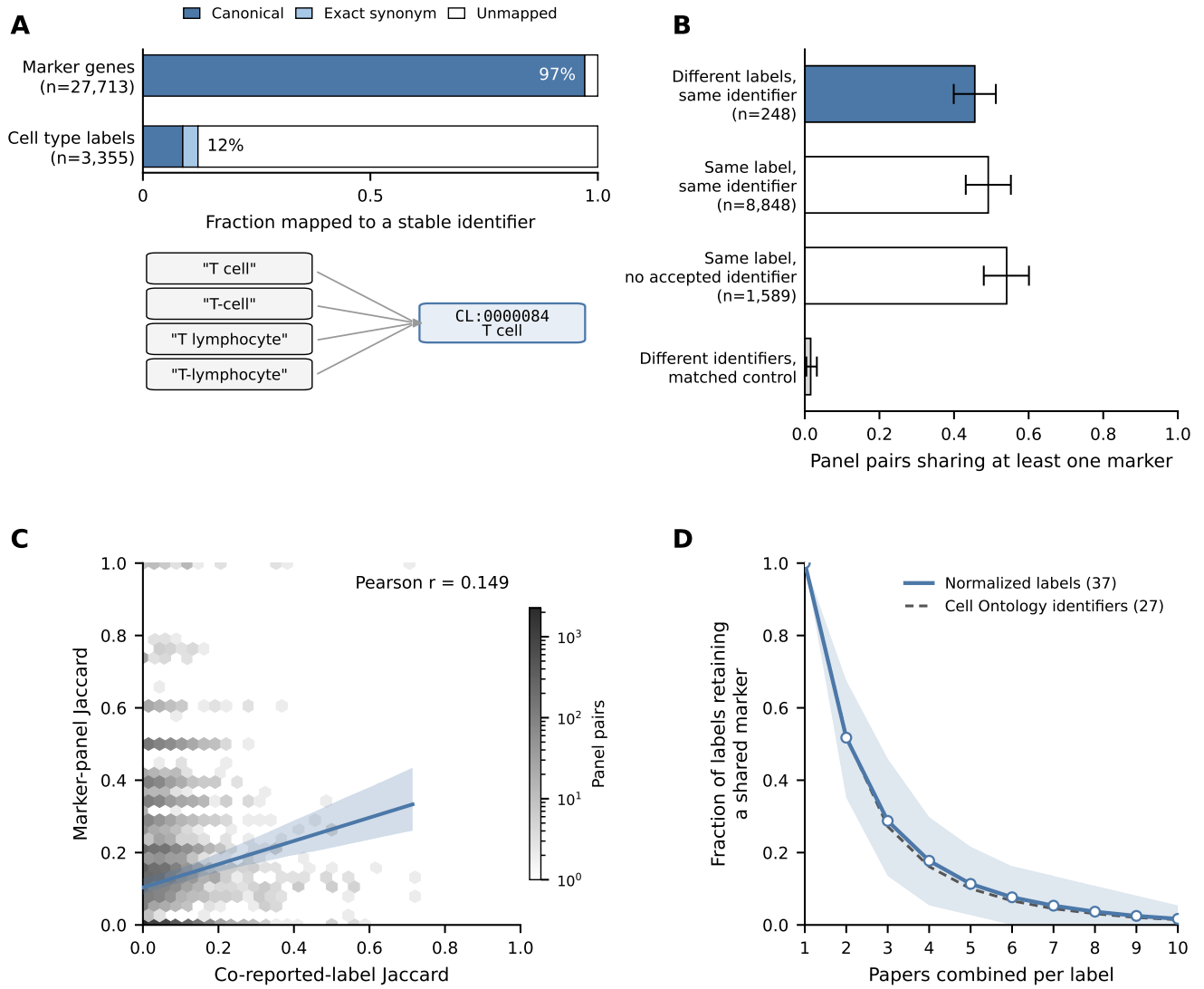


Figure 3: Stable identifiers connect markers across papers, but reported marker panels remain variable.

A. Fraction of reported marker genes and cell type labels mapped to stable identifiers. Cell type labels are split by canonical mapping versus synonym mapping. The illustration shows an example of cell type labels that map to one identifier. **B.** Fraction of marker gene panel pairs sharing at least one marker. A panel is the set of at least three mapped positive genes reported for one cell type in one paper. The white and gray bars use samples matched to the blue bar, with n giving each pool size (Methods); whiskers are 95% intervals. **C.** Marker-panel versus co-reported-label Jaccard for the 10,437 same-label panel pairs shaded by log density. The line and band are a least-squares fit with a 95% bootstrap interval. **D.** Fraction of labels retaining a shared marker as panels are intersected across papers. The band spans the 2.5th–97.5th percentiles; the dashed line uses the 27 accepted Cell Ontology identifiers.

group t , and comparison group of cells h , let $M_{s,p}(t, h)$ be the genes called as markers for t against h . A single marker gene for a set of comparison groups C must pass every comparison,

$$M_{s,p}(t; C) = \bigcap_{h \in C} M_{s,p}(t, h), \quad (1)$$

while a marker gene panel P needs only to cover every comparison: $P \cap M_{s,p}(t, h) \neq \emptyset$ for each $h \in C$. A cell type is then a group of cells identified by those markers

(subject to the stated comparisons).

We formalized these definitions and their consequences in the Supplementary Note. The formalization proves that marker gene sets are intersections of pairwise marker sets, marker gene panels are hitting sets over the comparisons, and adding a comparison can only remove single-gene markers or add requirements to a panel.²⁷

We further prove that one-vs-all DE testing is not mathematically equivalent to testing against each group pairwise. A pooled one-vs-rest statistic averages the

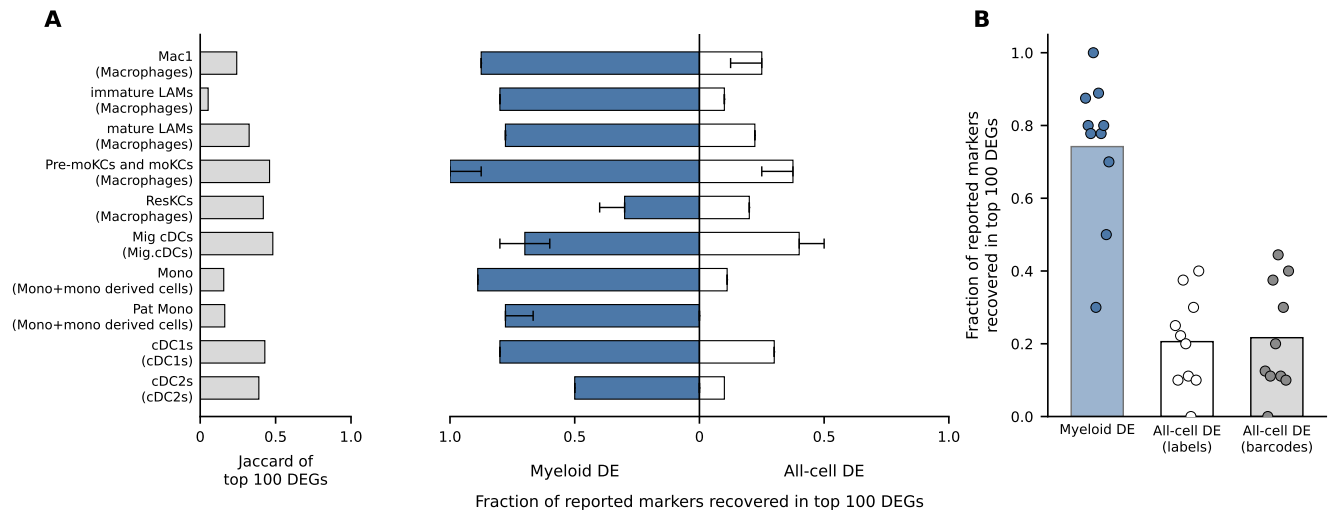


Figure 4: Marker recovery changes when the comparison groups change. The same liver myeloid cells were analyzed in a myeloid analysis and an all-cell liver analysis. **A.** For each myeloid cell type, the Jaccard similarity between the top 100 DEGs of the two analyses (left) and the fraction of reported markers recovered in the top 100 DEGs of each analysis (right). Whiskers are 95% intervals from 50 cell-type-stratified subsampling replicates and can exclude the observed value. **B.** Mean recovery for the myeloid analysis and for the all-cell analysis with target groups defined by the study’s labels or by the exact myeloid barcodes. Bars are means across the ten cell types and points are cell types. Per-cell-type values shift between the two all-cell definitions, but the means agree.

non-target groups, so a gene can pass the pooled test while failing the comparison against one specific group.²⁸ The formal result also explains hierarchical markers. A marker that passes a broad set of comparisons remains valid when the set is restricted, but a marker found among closely related cell groups need not identify the broader parent group, as illustrated in the CAP analysis [Domcke and Shendure, 2023, Beltz et al., 2026].

The formalization also clarifies the role of stable identifiers for defining marker genes. Stable identifiers make t and h joinable across papers, but a shared label or identifier does not imply a shared marker panel (Supplementary Figure S2). The utility of `mrkr` is to normalize labels and assign the stable identifiers that make this join possible.

3 Discussion

Marker genes are the practical endpoint of a single-cell RNA-seq experiment. They are the discriminating features that attach cell type labels to clusters, and they are expected to transfer between studies. Often they do not, because cell type and gene labels vary across papers, because reported markers mix data with the authors’ prior knowledge and judgment and cannot be fully reconstructed from the data alone, and because markers depend on their cellular context, which the flat cell type–gene format discards.

Large language models can address the first two problems. In our benchmark, extraction was roughly fourfold more accurate than generating markers from cell type

names or selecting them from ranked DE tables. `mrkr` uses an LLM to curate normalized marker labels into an `.onto` file, and each extracted marker keeps the supporting sentence that reported the association. Deterministic mapping to stable identifiers then joins associations reported under different names across papers, at a scale no manual curation can match. This solves the naming problem, the first step toward reuse. We release the resulting corpus of 979 validated `.onto` files, their validation metadata, and a derived SQLite database.²⁹

The third problem, tracking the cellular context, is harder. Even with stable identifiers, markers are context dependent. Shared markers for the same cell type vanished as evidence accumulated, and changing only the comparison group of cells changed which reported markers were recovered. An LLM can extract one part of the cellular context, the comparison groups, but only when a paper states them, and most papers do not. Our curation cannot fix that. In that sense, the missing comparison groups are a sociological problem that manifests as a reproducibility issue. The fix is to deploy `mrkr` at authoring time, while the cellular context is still known, recording the reported label, normalized label, stable identifier, supporting sentence, and comparison group of cells together in an `.onto` file.

Knowing the cellular context and reusing stable identifiers are steps toward resolving the debate over the definition of a cell type [Trapnell, 2015, Osumi-Sutherland et al., 2021]. All sides of that debate correctly point to the importance of measurements for defining cell types, but the debate has lacked a pragmatic resolution. We

propose that marker genes, recorded with their cellular context, are that resolution. We prove that a marker gene is valid relative to the comparisons that produced it and that a marker panel covers those comparisons. This definition can be shared because it names exactly what must be recorded for a marker to be reused. While these

reporting details cannot remove assay variation or supply cell groups that were never observed, they make the comparison behind each marker observable and auditable. They prevent missing comparison information from being confused with those limitations. This is what `mrkr` and `.onto` fix.

A Supplementary Material

Table S1: LLMarkers benchmark. The number of unique cell types (CTs), genes (Ensembl IDs), and cell type–gene associations Overall, and across Text, Image, and DEG tables. The %DEG reports the percentage of associations found in the study’s DEG lists. Total rows are unique counts across studies and are smaller than column sums.

Dataset	Overall			Text				Image				DEG
	CTs	Genes	Pairs	CTs	Genes	Pairs	DEG%	CTs	Genes	Pairs	DEG%	Pairs
Adipose (Emont)	45	105	346	20	23	30	80	44	104	344	39	43,471
Adipose (Hildreth)	31	141	234	21	53	66	92	31	124	207	96	46,357
Bone (He)	27	75	370	26	69	98	53	24	29	309	13	5,382
Eye (Gautam)	32	94	251	12	22	24	88	28	82	235	49	24,998
Lung (Adams)	20	124	135	6	33	33	97	20	99	109	97	34,760
Ovary (Wagner)	7	62	58	7	46	37	89	6	54	41	80	14,357
Testis (Shamis)	19	125	171	15	91	123	78	11	59	64	94	9,663
Total	168	641	1,560	100	305	409	78	151	495	1,304	53	169,687

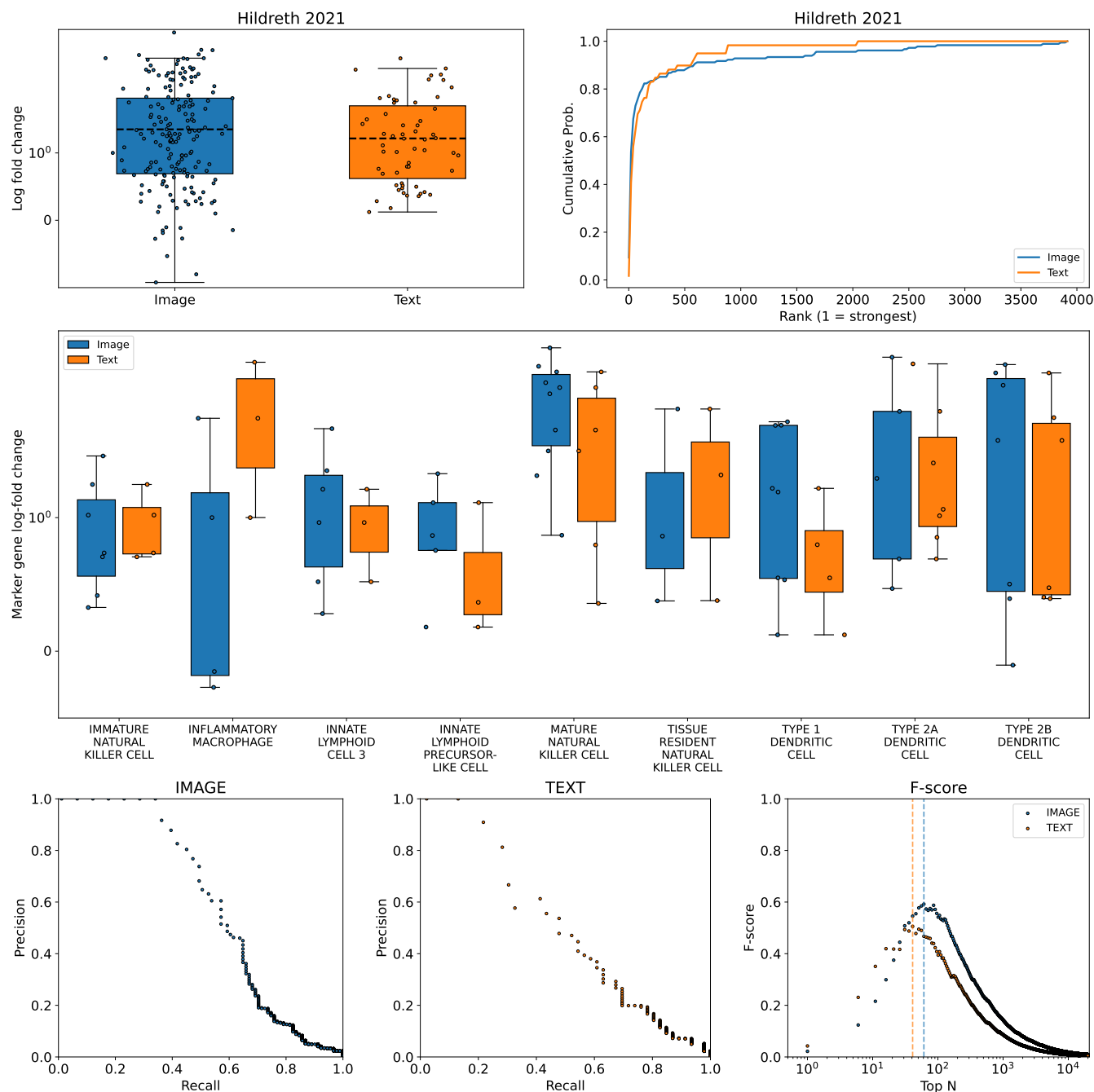


Figure S1: Reported marker strength and recovery vary by cell type in Hildreth et al. [2021]. Top: log-fold-change distributions and cumulative DEG-rank distributions for markers reported in figures (blue) and text (orange). Middle: log-fold-change distributions stratified by cell type and source. Bottom: precision-recall curves and F1 across top- N DEG thresholds.

Table S2: LLM marker gene curation. All runs used Claude (claude-opus-4-6). Precision (P), recall (R), and F1 were computed against positive human text annotations. *N*: unique mapped cell type–gene associations produced. %DEG: produced triples found in the matching DEG table. **(a)** Generation from cell type names. Pred. Acc.: cell types identified from anonymized gene lists. **(b)** Selection from ranked DEGs. Best *N*: input depth maximizing pair F1. **(c)** Joint extraction from manuscript text and the catalog of candidate DEG sources. Pair scores require the exact normalized cell type and gene; triple scores also require the exact supporting `data_id`. Retained associations passed current `mrkr` exact-source validation. Bottom rows show means across datasets.

(a) Generation								
Dataset	Pred. %	<i>N</i>	P	R	F1	Cost (\$)	Time (min)	
Adipose (Emont)	13	317	0.03	0.37	0.06	0.49	3.2	
Adipose (Hildreth)	10	249	0.13	0.48	0.20	0.41	2.8	
Bone (He)	4	203	0.14	0.30	0.19	0.34	2.3	
Eye (Gautam)	13	243	0.04	0.37	0.07	0.39	2.6	
Lung (Adams)	40	165	0.07	0.48	0.12	0.30	2.0	
Ovary (Wagner)	29	69	0.26	0.33	0.29	0.12	0.8	
Testis (Shamis)	16	144	0.20	0.24	0.22	0.25	1.7	
Mean	18	199	0.13	0.37	0.17	0.33	2.2	

(b) Selection								
Dataset	Best <i>N</i>	<i>N</i>	%DEG	P	R	F1	Cost (\$)	Time (min)
Adipose (Emont)	200	371	97	0.03	0.33	0.05	1.53	4.4
Adipose (Hildreth)	400	333	98	0.13	0.67	0.22	0.50	4.2
Bone (He)	300	293	97	0.13	0.38	0.19	0.95	3.4
Eye (Gautam)	20	218	95	0.06	0.44	0.10	0.43	2.7
Lung (Adams)	200	349	97	0.03	0.43	0.05	1.24	3.9
Ovary (Wagner)	200	110	100	0.17	0.35	0.23	0.35	1.3
Testis (Shamis)	300	227	99	0.19	0.36	0.25	0.93	2.8
Mean		272	98	0.11	0.42	0.16	0.85	3.2

(c) Extraction									
Dataset	Sources	Claims	Gene terms	Cell type–gene			+ data ID		
				P	R	F1	P	R	F1
Adipose (Emont)	6	19	26	0.73	0.63	0.68	0.65	0.57	0.61
Adipose (Hildreth)	4	17	47	1.00	0.71	0.83	1.00	0.71	0.83
Bone (He)	5	34	57	0.78	0.40	0.53	0.73	0.39	0.51
Eye (Gautam)	7	24	33	0.48	0.59	0.53	0.48	0.57	0.52
Lung (Adams)	8	12	44	0.52	1.00	0.69	0.41	0.78	0.54
Ovary (Wagner)	3	10	35	0.81	0.45	0.58	0.65	0.59	0.62
Testis (Shamis)	6	36	69	0.89	0.48	0.62	0.89	0.48	0.62
Mean		22	44	0.75	0.61	0.64	0.69	0.59	0.61

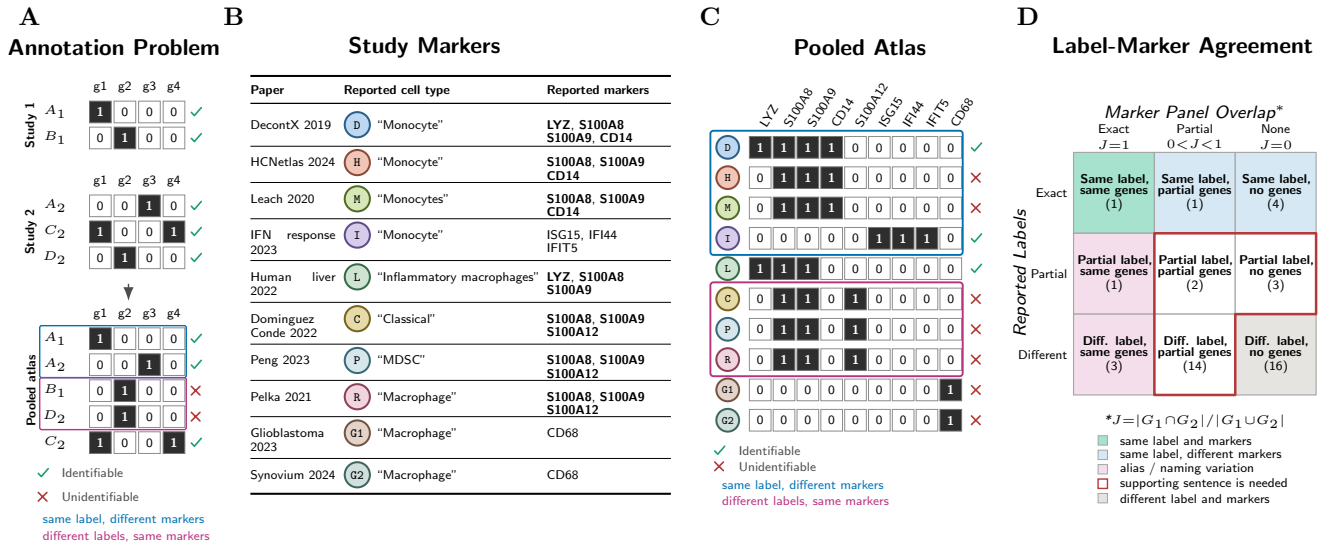


Figure S2: Marker genes and cell type labels provide different information. **A.** Binary marker matrices illustrate how a marker selected within one study can fail after studies are combined. Colored outlines indicate identical labels with different markers and different labels with identical markers. **B.** Myeloid examples curated from the corpus; bold genes are shared by at least two reported cell groups. **C.** Binary marker matrix for the examples in **B.** **D.** Relationship between label similarity and marker-panel similarity among the displayed examples. Marker overlap is Jaccard similarity, $J = |G_1 \cap G_2| / |G_1 \cup G_2|$.

B Methods

B.1 Benchmark construction and verification

The LLMarkers benchmark was assembled by manual curation of seven single-cell RNA-seq studies spanning six tissues: **adipose** [Emont et al., 2022, Hildreth et al., 2021], **bone** [He et al., 2021], **eye** [Gautam et al., 2021], **lung** [Adams et al., 2020], **ovary** [Wagner et al., 2020], and **testis** [Shami et al., 2020]. The benchmark was designed as a carefully linked evaluation set rather than a comprehensive survey of single-cell reporting styles. For each study, a curator read the manuscript and recorded each marker gene–cell type association, its verbatim sentence or figure evidence, source type, linked differential-expression table when available, organism, and Ensembl gene identifier. We preserved the original flat records and converted them losslessly into separate text statements, image statements, and differential-expression tables. Repeated source rows remain traceable through record identifiers rather than being discarded. Ten mouse records from the Adams study remain in the all-organism benchmark totals; the current extraction evaluation is restricted to human records.³⁴

Each study’s supplementary tables were parsed to extract differential expression gene lists. Header detection was performed automatically by scanning the first 10 rows for a row containing expected column names (e.g., gene, p-value, avg_log2FC). The resulting DEG lists were stored in the same evidence schema as human annotations, enabling direct comparison.³⁵

We verified the converted benchmark against both the source records and manuscript text. Every source row round-tripped exactly and every source-file hash was unchanged. Of 166 primary text evidence claims, 158 (95%) were anchored to exact manuscript offsets; the remaining eight are retained with an explicit unanchored status rather than treated as verified text.³⁶

B.2 Marker strength analysis

To quantify where human-curated marker genes fall in the differential expression ranking, we matched each marker to its corresponding entry in the dataset’s DEG list. For human-curated markers, matching was performed on the four-part key (dataset, data_id, group_name, feature_id), requiring exact agreement on the cell type, gene, and supplementary data source. Markers that could not be matched—because the gene was absent from the DEG list or lacked an Ensembl ID—were excluded from this analysis.³⁷

For each matched marker, we recorded its rank within the cell type’s DEG list (rank 1 = highest log-fold change) and its log-fold change. We computed the fraction of markers falling within the top N DEGs at every threshold N from 1 to 200, reported the values at $N = 50$ and $N = 100$, and separately tabulated recovery for text-sourced and image-sourced markers.³⁸

To estimate the optimal DEG list depth for recovering each marker set, we computed precision–recall curves by varying a rank threshold N from 1 to 20,000. At each threshold, we defined the top N DEGs as predicted markers and computed precision, recall, and the F-score. The optimal N was defined as the threshold that maximized the F-score. This analysis was performed separately for each dataset and marker source (text or image).³⁹

B.3 LLM generation and extraction

We evaluated three approaches to LLM-based marker gene curation across the seven benchmark datasets. All experiments used Claude (claude-opus-4-6) with temperature 0 and a maximum output length of 32,000 tokens; we did not benchmark model-to-model variation. Costs were computed from the model’s API pricing (\$5 per million input tokens, \$25 per million output tokens for claude-opus-4-6).⁴⁰

In the generation experiment (celltypes-to-genes), we gave the model the list of cell type names from each dataset’s human-curated marker set and asked it to generate 5–15 well-established marker genes per cell type. The model received only the species and cell type names—no manuscript text, figures, or supplementary data. In a second generation experiment (genes-to-celltypes), we extracted the gene list for each cell type from the human-curated set, replaced the cell type names with anonymous labels (Group 1, Group 2, ...), and asked the model to predict the cell type for each group. We measured prediction accuracy as the fraction of exact string matches between predicted and original cell type names. Non-exact predictions were counted as related when they showed substring containment or shared label tokens; remaining predictions were counted as having no lexical relation.⁴¹

In the selection experiment, we parsed each dataset’s DEG tables, ranked genes per cell type by absolute log fold change, and presented the top N genes (with log fold change and adjusted p -value) to the model, asking it to select which are true markers. We swept $N \in \{10, 20, 30, 40, 50, 100, 200, 300, 400, 500\}$ for each dataset and reported the N that maximized pair-level F1. As a control for whether cell type identity aids selection, we repeated the full sweep with cell type names replaced by anonymous cluster labels (Cluster 1, Cluster 2, ...), preserving the gene lists and statistics but removing cell type identity from the names.⁴²

To quantify the best possible selection performance, we computed a DEG-constrained max F1 score at each N . Let H be the set of human-curated marker pairs and D_N be the set of marker pairs available in the top- N DEG lists shown to the model. The recoverable set is $R_N = H \cap D_N$. Any selector restricted to D_N cannot recover pairs outside R_N . The maximum achievable pair-level F1 is therefore

$$\text{F1}_{\max}(N) = \frac{2|R_N|}{|H| + |R_N|}, \quad (2)$$

which is attained by selecting exactly the recoverable set R_N (precision = 1). In **Figure 2C**, the selection upper bound is plotted as open circles at each dataset’s selected operating point, using that same dataset-specific N ; the dashed line marks their mean.

In the joint extraction experiment, Claude received the full manuscript and a catalog mapping each candidate `data_id` to the cell type labels present in that paper’s DEG sources. For every reported human marker association, the model returned the cell type, gene, verbatim supporting text, and one candidate `data_id`, or null when no source was supported. The model did not receive the human annotations. When it returned null, an exact target-label match supplied the narrowest candidate source; this affected five of 152 retained claims. We converted the output to the current `mrkr` claim structure and coalesced records with the same source text and target into marker panels.⁴³

The extraction evaluation used positive human text annotations. Validation required every retained claim to have an exact manuscript offset and every retained marker gene to occur explicitly in that source text. Unsupported genes and claims with ambiguous source offsets were excluded. Gene labels were mapped programmatically with the packaged `gmap.txt`. We computed macro-averaged precision, recall, and F1 across studies for exact normalized cell type–gene pairs and exact `data_id`–cell type–gene triples. The same extracted terms were used for both scores.⁴⁴ Validation retained 152 associations listing 311 mapped marker genes and excluded five as unsupported or ambiguous.⁴⁵

The joint experiment saved each raw model response before validation. Every derived artifact records hashes for the prompt, manuscript, human benchmark, DEG catalog, and gene map. The run is therefore resumable and fails if an input changes.⁴⁶

B.4 Paper corpora

We selected bioRxiv preprints from a local mirror of 428,741 MECA archives downloaded from the bioRxiv source repository in January 2025. Each archive contains JATS XML and structured metadata.⁴⁷ We required a CC-BY or CC0 license, terms that indicated human relevance, and terms related to cell types, single-cell RNA sequencing, or marker genes. We kept the most recent version for each DOI. These filters produced 504 bioRxiv manuscripts.⁴⁸ JATS XML was converted to Markdown while preserving section structure.

We assembled the second corpus from publications linked to Human Cell Atlas projects. Project metadata supplied publication titles, URLs, and DOIs. We deduplicated records by DOI and then by normalized title. Of 515 linked publication records, 475 had manuscript text that could be normalized to Markdown and processed.⁴⁹

B.5 Label normalization and identifier grounding

The corpus source manifest lists one paper identifier, paper source (bioRxiv or HCA), organism, manuscript path, and source hash per paper. We processed all 979 manuscripts with `mrkr` version 0.4.0 at the pinned source revision in `analysis/mrkr.lock`. Extraction used Claude (claude-opus-4-6). The model received the complete Markdown manuscript and the organism but could not assign database identifiers.

For each paper, `mrkr extract` returned exact source text, normalized labels, and term offsets. `mrkr ground` mapped human genes with the packaged gene map, organisms to NCBI Taxonomy, cell types and comparisons to Cell Ontology, and tissues to UBERON. `mrkr validate` then checked the source hash, source offsets, explicit term offsets, required term counts, gene direction, and grounding metadata.⁵⁰ Canonical Ensembl GRCh38 release 114 symbols take priority when a gene alias is ambiguous. Each saved `.onto` file records the model and response identifier, prompt digest, gene-map digest, ontology queries, selected identifiers, and validation state. The archive stores the raw model responses separately.

Some initial outputs contained individual associations or gene terms that failed exact-source validation. We repaired these files without another model call. The repair removed only the failing association or conflicting gene term, recorded every removal, and reran grounding and validation.⁵¹ Papers with no retained association remained in the corpus as valid empty results. We built the analysis database from the final ontology manifest. The database stores paper metadata, reported cell type–gene associations, exact source text and offsets, normalized terms, identifiers, directions, and grounding state. Ontology hierarchy expansion was disabled for the primary analysis.

B.6 Cross-paper links by labels and identifiers

We analyzed mapped positive marker genes separately from mapped negative marker genes. A reported marker gene panel is the set of mapped positive genes assigned to one normalized cell type label in one paper. Labels were normalized with Unicode NFKC, case folding, replacement of non-word characters with spaces, and whitespace normalization. The primary analysis used exact normalized-label equality. It did not use substring, token-overlap, or semantic label matching. We retained panels with at least three mapped genes.

We grouped source records by normalized article title and excluded comparisons within the same article. For every cell type label reported in at least two articles, we compared all retained paper pairs. Marker similarity was the Jaccard similarity between the two sets of Ensembl gene identifiers. We also recorded whether the sets shared any gene and the fraction of each set recovered by the other.

For each pair of papers that used the same cell type label, we sampled one pair with different labels. Control panels matched the original paper source and the base-2 marker-count bin. We required different article keys and repeated the control sampling with 25 fixed random seeds.⁵² Pair-weighted estimates give each paper pair equal weight. Label-balanced estimates first average the paper pairs for each label and then give each recurring label equal weight. We used 5,000 label-bootstrap replicates for the 95% intervals among papers that used the same label. Control intervals are the 2.5th and 97.5th percentiles across the 25 matched draws.

To measure cumulative marker-panel intersection, we fixed the cohort to normalized labels reported in at least ten papers. In each of 1,000 bootstrap replicates, we resampled labels with replacement, randomly ordered the paper-level panels for each selected label, and intersected the first one through ten panels. At each step, we recorded whether the intersection contained at least one marker. The plotted estimate is the mean fraction of labels with a nonempty intersection. Intervals are the 2.5th and 97.5th percentiles across replicates. We repeated the calculation for accepted Cell Ontology identifiers and obtained similar results. Separate sensitivity analyses repeated the pairwise and all-paper intersection summaries with minimum panel sizes of one, three, five, and ten genes.

To measure co-reported labels, we removed the shared cell type label from each paper's label set and computed Jaccard similarity between the remaining sets. We then compared this value with marker-panel Jaccard similarity. The fitted line is an ordinary least-squares fit. Its band is a 95% interval from 1,000 bootstrap replicates that resampled normalized target labels as clusters. Pearson correlation summarizes the unbinned association. Co-reported labels describe the labels in a paper. They do not reconstruct the cell-level comparison.

We performed a separate ontology analysis with Cell Ontology release v2026-06-08. A reported label was eligible only when it equaled the canonical label or an official exact synonym for its mapped term. We excluded broad, narrow, related, and matches supported only by full-text coverage. Identifier-recovered pairs shared an accepted Cell Ontology identifier but used different normalized labels. For every such pair, we sampled three references matched by paper source and the unordered pair of base-2 marker-count bins: a same-label pair with the same accepted identifier, a same-label pair without an accepted identifier, and a pair with different identifiers. Because grounding is deterministic from the normalized label, no same-label pair had two different accepted identifiers. We gave each identifier-recovered pair equal weight. Intervals are the 2.5th and 97.5th percentiles from 1,000 pair-bootstrap and matched-sampling replicates. An ontology-term-balanced analysis is retained as a sensitivity analysis.⁵³

The matrices in **Supplementary Figure S2** are schematic. The myeloid examples are validated associations from the corpus, selected to illustrate relations between reported labels and marker gene panels. No corpus estimate is computed from these examples.⁵⁴

B.7 Cell Annotation Platform expression analysis

We used CAP dataset 1440 as the liver myeloid analysis and CAP dataset 1437 as the broader all-cell liver analysis. The files are nested, not independent. All 40,821 cell barcodes in dataset 1440 occur in the 167,598-cell all-cell file, which contains 17 author cell type labels.⁵⁵ We loaded each file with `anndata.read_h5ad`, used the `user_provided` expression layer when present, and used `author_cell_type` as the label. We mapped each myeloid label to its broader author label in the all-cell analysis. The two annotations disagree for some cells: 83.8% of the myeloid cells carry their mapped broad label in the all-cell analysis. As a control, we recomputed the all-cell one-vs-rest rankings with each target group defined by the exact barcodes of the corresponding myeloid label instead of the study's all-cell labels; mean top-100 recovery was 21.7%, matching the label-based 21%.⁵⁶

For each label, we ranked all genes by the difference in mean expression between the cell group of interest and the remaining cells, divided by its standard error. Reported markers came from the CAP marker JSON and were restricted to genes mapped to human Ensembl identifiers. Top-100 recovery is the fraction of reported markers among the first 100 ranked genes. Differential-expression stability is the Jaccard similarity between the myeloid and all-cell top-100 sets. Marker retrieval F1 treats the reported markers as positives and reports the maximum F1 over rank cutoffs.

For uncertainty in top-100 recovery, we performed 50 cell-type-stratified subsampling replicates. Each replicate sampled 80% of cells within each author label, with at most 10,000 cells per label. Figure 4A shows the 2.5th and 97.5th percentiles.

B.8 Lean formalization

We encoded the formal results in Lean 4 under `analysis/formal`. The binary model records studies, partitions, cell groups, pairwise comparisons, marker genes, and marker gene panels. The score model also records real-valued marker scores, thresholds, weighted backgrounds, and per-group mean summaries.⁵⁷ The alignment between the Supplementary Note and the Lean declarations is curated in the `span` format.

The Lean development proves the marker-family intersection theorem, monotonicity under restriction of the comparison groups, local-to-global and hierarchy results, equivalence between the all-pair conjunction and the binary marker-family definition, marker panels as hitting sets, panel monotonicity, weighted-background identities, score-thresholded results, and examples in which a cell type–gene projection loses comparison information.⁵⁸ The Supplementary Note separates the intersection of pairwise tests from a pooled one-vs-rest statistic.

B.9 Data and code availability

Analysis code, the human benchmark, validated corpus archive, derived result tables, figure code, and Lean formalization are available at <https://github.com/sbooeshaghi/llmarkers>. The `mrkr` tool is available at <https://github.com/sbooeshaghi/mrkr>. The corpus archive contains the 979 validated `.onto` files, source and ontology manifests, raw model responses, repair reports, validation metadata, and the derived SQLite database. The original manuscripts are not redistributed when their licenses do not allow it. Each source manifest records the paper identifier and expected manuscript hash so that licensed source text can be restored and validated.

C Acknowledgements

We thank Ben Johnson, Josh Liang, and Maggie Chasse for fruitful discussions about cell types. We also thank Michal Rozenwald, Jase Gehring, Jeremy Marcus, and Kevin An for feedback on a first draft of these results, as well as attendees at the Cancer Genetics and Epigenetics Gordon Research Seminar who commented on an earlier version of this work. We thank Luca Pinello for suggesting the formalization of local and global marker genes and Daniel Lewinsohn for suggesting that we extend the formalization to marker panels. We also thank all of humanity for their contributions to the training data of the large language models that this work relied on.

D Author contributions

A.S.B. conceived of the study, and compiled the first draft of the benchmark. N.A., along with A.S.B., completed the benchmark and performed initial analyses for verifying correctness of the **LLMarkers**. A.W. suggested expanding the analysis to the bioRxiv and Human Cell Atlas-linked corpora. A.S.B. performed the remaining analysis, along with Claude Opus 4.6, and wrote the manuscript. All authors met and discussed results and reviewed the manuscript.

E Disclosures

All em dashes were placed, in the text, by the author’s own hands. Generative language models were used by the authors to check for grammatical errors in the text, fix bugs in code, and debug L^AT_EX issues.

Claim evidence registry

#	ID	Claim	Sources
1	fig-mrkr-overview	<i>Figure/table source record</i>	docs/paper/biorxiv/tex-figures/fig1_mrkr_overview_body.tex README.md
2	benchmark-summary	We manually curated 2,528 reported cell type–gene associations from seven scRNA-seq studies spanning six tissues, together with each study’s differential-expression (DE) tables, and call this benchmark LLMarkers . The benchmark covers 168 cell types, 641 genes, and 1,560 unique cell type–gene associations (Supplementary Table S1).	analysis/artifacts/benchmark_evidence_v1/reconciliation.json analysis/benchmark_verification.ipynb
3	benchmark-modality	Most markers were reported only in figures: 1,151 of the 1,560 associations (74%), with 256 (16%) only in text and 153 (10%) in both (Figure 2A).	analysis/benchmark_verification.ipynb
4	fig-marker-selection	<i>Figure/table source record</i>	analysis/build_fig2_marker_recovery.py analysis/artifacts/fig2_modality.tsv analysis/artifacts/fig2_deg_presence.tsv
5	fig-llm-curation	<i>Figure/table source record</i>	analysis/build_fig2_marker_recovery.py analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv analysis/evaluate_mrkr_joint_deg_extraction.py
6	benchmark-deg-overlap	The fraction of reported markers present in a study’s DE tables ranged from 18% in the embryonic skeletal development study [He et al., 2021] to 97% in the idiopathic pulmonary fibrosis study [Adams et al., 2020] (Figure 2A). In the Emont adipose study, the authors report <i>PDGFRA</i> as a marker of subpopulations hASPC1 through hASPC6 , but the DE table for those subpopulations includes it only for hASPC1 .	analysis/benchmark_verification.ipynb
7	benchmark-sign-contradictions	Among 544 marker gene associations found in at least two DE tables from the same study, 46 (8.5%) were upregulated in one table and downregulated in the other.	analysis/benchmark_verification.ipynb
8	expert-ranks	Among markers reported in text, 53% appeared in the top 50 differentially expressed genes (DEGs); among markers reported in figures, 50%. The top 100 DEGs contained 66% and 62%.	analysis/artifacts/fig2_rank_recovery_summary.tsv analysis/build_fig2_marker_recovery.py analysis/lfc_comparison.ipynb
9	expert-optimal-n	Averaged across studies, F-score peaked at only 0.41 for markers reported in text and 0.42 for markers reported in figures, near cutoffs of 30 and 40 genes (Figure 2B), and per-study optimal cutoffs ranged from 21 to 186 genes.	analysis/artifacts/fig2_rank_recovery_summary.tsv analysis/artifacts/fig2_optimal_deg_cutoff.tsv analysis/build_fig2_marker_recovery.py
10	gen-f1	Generation from cell type names reached a mean cell type–gene F1 of 0.17 against the markers reported in text.	analysis/artifacts/fig2_llm_recovery.tsv analysis/build_fig2_marker_recovery.py
11	sel-f1	Selection from ranked DE tables reached 0.16 at the best input depth for each study.	analysis/artifacts/fig2_llm_recovery.tsv analysis/build_fig2_marker_recovery.py

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#	ID	Claim	Sources
12	ext-text-f1	Extraction reached a mean F1 of 0.64, and 0.61 when the model also identified the DE table supporting each association.	analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_summary.json analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv
13	mrkr-record-contract	Each retained association links the supporting sentence to one organism, one cell type label, at least one marker gene, the marker direction, and any tissue or comparison group of cells named in the paper. Validation checks that the supporting sentence and every extracted term occur exactly in the manuscript (Methods). The LLM extracts named cell types and genes and produces normalized labels that are mapped, deterministically, to stable identifiers from Ensembl, NCBI Taxonomy, Cell Ontology, and UBERON.	analysis/mine_corpus.py analysis/build_claim_db.py README.md
14	corpus-source-grounded-summary	Of the 979 papers, 721 reported at least one cell type-gene association, totaling 11,336 associations of 27,713 marker genes. 586 of the 11,336 associations (5.2%) named a comparison group of cells.	analysis/artifacts/mrkr_corpus_analysis_v1/corpus_summary.tsv analysis/artifacts/mrkr_corpus_analysis_v1/term_coverage.tsv analysis/artifacts/mrkr_corpus_analysis_v1/report.md
15	corpus-gene-mapping	Of the 27,713 reported marker genes, 26,919 (97%) mapped to an Ensembl identifier.	analysis/artifacts/mrkr_corpus_analysis_v1/corpus_summary.tsv analysis/artifacts/mrkr_corpus_analysis_v1/term_coverage.tsv
16	corpus-identifier-linkage	Of 3,355 distinct cell type labels, 294 matched a canonical label, 112 matched an exact synonym, and the remaining 2,949 (88%) matched neither. The 406 accepted labels (12%) covered 363 identifiers.	analysis/artifacts/mrkr_corpus_analysis_v1/cell_ontology_label_audit.tsv
17	corpus-label-collapse	“T cell,” “T-cell,” “T lymphocyte,” and “T-lymphocyte” all mapped to CL:0000084. Overall, 324 of the 363 identifiers received one reported label, 37 received two, and two identifiers received four.	analysis/artifacts/mrkr_corpus_analysis_v1/cell_ontology_label_audit.tsv
18	corpus-analysis-ready	Keeping panels with at least three genes produced 3,454 panels and 10,437 panel pairs.	analysis/artifacts/mrkr_corpus_analysis_v1/label_profiles_analysis.tsv analysis/artifacts/mrkr_corpus_analysis_v1/same_label_pairs.tsv
19	corpus-label-signal	Twelve identifiers connected 248 panel pairs with different normalized labels, such as “plasma B cell” and “plasma cell,” and 113 of the 248 (45.6%) shared at least one marker. Panel pairs whose labels matched exactly shared a marker at a similar rate (49.2% with a shared identifier, 54.1% without, after matching by paper source and panel size). Panel pairs with different identifiers shared a marker 1.6% of the time.	analysis/artifacts/mrkr_corpus_analysis_v1/same_label_pairs.tsv analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_pairs.tsv analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_matched_summary.tsv
20	corpus-coreported-context	Across the 10,437 panel pairs, marker sharing was weakly positively associated with the similarity of the co-reported labels (Pearson $r = 0.149$).	analysis/artifacts/mrkr_corpus_analysis_v1/same_label_pairs.tsv analysis/artifacts/mrkr_corpus_analysis_v1/coreported_context_metrics.tsv
21	corpus-marker-accumulation	After two papers, an estimated 51.8% of labels retained at least one shared marker. The fraction fell to 28.8% after three papers, 11.3% after five, and 1.7% after ten. Repeating the analysis with 27 accepted Cell Ontology identifiers gave nearly identical results.	analysis/artifacts/mrkr_corpus_analysis_v1/label_intersection_accumulation.tsv analysis/artifacts/mrkr_corpus_analysis_v1/ontology_intersection_accumulation.tsv

Continued on next page

#	ID	Claim	Sources
22	fig-corpus-marker-reuse	<i>Figure/table source record</i>	analysis/build_fig3_corpus_reuse.py analysis/analyze_mrkr_corpus.py analysis/artifacts/mrkr_corpus_analysis_v1/term_coverage.tsv analysis/artifacts/mrkr_corpus_analysis_v1/cell_ontology_label_audit.tsv analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_pairs.tsv analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_matched_summary.tsv analysis/artifacts/mrkr_corpus_analysis_v1/same_label_pairs.tsv analysis/artifacts/mrkr_corpus_analysis_v1/coreported_context_metrics.tsv analysis/artifacts/mrkr_corpus_analysis_v1/coreported_context_fit.tsv analysis/artifacts/mrkr_corpus_analysis_v1/label_intersection_accumulation.tsv analysis/artifacts/mrkr_corpus_analysis_v1/ontology_intersection_accumulation.tsv
23	fig-cap-partition	<i>Figure/table source record</i>	analysis/build_fig4_cap_partition.py analysis/build_cap_liver_local_global_deg.py analysis/artifacts/cap_liver_barcode_constant_recovery.tsv analysis/build_cap_liver_barcode_constant_deg.py analysis/artifacts/cap_liver_local_global_deg_summary.tsv analysis/artifacts/cap_liver_local_global_recovery_subsampling_summary.tsv
24	cap-liver-local-global-de	Across the ten myeloid labels, one-vs-rest differential expression within the myeloid analysis recovered a mean of 74% of reported markers in the top 100 genes. The all-cell analysis recovered 21%. Recovery was higher in the myeloid analysis for every label.	analysis/artifacts/cap_liver_local_global_deg_summary.tsv analysis/artifacts/cap_liver_local_global_recovery_subsampling_summary.tsv analysis/build_cap_liver_local_global_deg.py
25	cap-liver-rank-overlap	The two analyses also produced different rankings: for every label, the top 100 DEGs of the two analyses overlapped by less than half (Jaccard 0.05–0.48; Figure 4A).	analysis/artifacts/cap_liver_local_global_deg_summary.tsv
26	cap-liver-barcode-constant	Defining each all-cell target group by the exact barcodes of its myeloid cells, rather than by the study’s all-cell labels, gave the same result: a mean recovery of 22% (Figure 4B).	analysis/artifacts/cap_liver_barcode_constant_recovery.tsv analysis/build_cap_liver_barcode_constant_deg.py
27	formal-marker-definition	The formalization proves that marker gene sets are intersections of pairwise marker sets, marker gene panels are hitting sets over the comparisons, and adding a comparison can only remove single-gene markers or add requirements to a panel.	analysis/formal/MarkerIdentifiability/Basic.lean analysis/formal/CLAIMS.md
28	formal-pooled-vs-pairwise	We further prove that one-vs-all DE testing is not mathematically equivalent to testing against each group pairwise. A pooled one-vs-rest statistic averages the non-target groups, so a gene can pass the pooled test while failing the comparison against one specific group.	analysis/formal/MarkerIdentifiability/Basic.lean analysis/formal/CLAIMS.md

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#	ID	Claim	Sources
29	corpus-resource-release	We release the resulting corpus of 979 validated .onto files, their validation metadata, and a derived SQLite database.	analysis/artifacts/mrkr_corpus_onto_v2.tar.gz.sha256 README.md
30	tab-benchmark	<i>Figure/table source record</i>	analysis/benchmark_verification.ipynb
31	fig-hildreth-detail	<i>Figure/table source record</i>	analysis/lfc_comparison.ipynb
32	tab-llm-curation	<i>Figure/table source record</i>	analysis/ext_vs_gen.ipynb analysis/selection_analysis.ipynb analysis/llm_extraction_eval.ipynb analysis/artifacts/fig2_llm_recovery.tsv analysis/artifacts/fig2_llm_recovery_current.tsv analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_pair_review.tsv analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_triple_review.tsv analysis/build_fig2_marker_recovery.py analysis/evaluate_mrkr_joint_deg_extraction.py
33	fig-paper-celltype-joint	<i>Figure/table source record</i>	docs/paper/biorxiv/tex-figures/fig_paper_celltype_joint_body.tex
34	methods-benchmark-schema	The LLMarkers benchmark was assembled by manual curation of seven single-cell RNA-seq studies spanning six tissues: adipose [Emont et al., 2022, Hildreth et al., 2021], bone [He et al., 2021], eye [Gautam et al., 2021], lung [Adams et al., 2020], ovary [Wagner et al., 2020], and testis [Shami et al., 2020]. The benchmark was designed as a carefully linked evaluation set rather than a comprehensive survey of single-cell reporting styles. For each study, a curator read the manuscript and recorded each marker gene–cell type association, its verbatim sentence or figure evidence, source type, linked differential-expression table when available, organism, and Ensembl gene identifier. We preserved the original flat records and converted them losslessly into separate text statements, image statements, and differential-expression tables. Repeated source rows remain traceable through record identifiers rather than being discarded. Ten mouse records from the Adams study remain in the all-organism benchmark totals; the current extraction evaluation is restricted to human records.	analysis/build_benchmark_evidence.py analysis/artifacts/benchmark_evidence_v1/reconciliation.json
35	methods-deg-parsing	Each study’s supplementary tables were parsed to extract differential expression gene lists. Header detection was performed automatically by scanning the first 10 rows for a row containing expected column names (e.g., gene, p_value, avg_log2FC). The resulting DEG lists were stored in the same evidence schema as human annotations, enabling direct comparison.	analysis/benchmark_verification.ipynb

Continued on next page

#	ID	Claim	Sources
36	methods-verification	Every source row round-tripped exactly and every source-file hash was unchanged. Of 166 primary text evidence claims, 158 (95%) were anchored to exact manuscript offsets; the remaining eight are retained with an explicit unanchored status rather than treated as verified text.	analysis/artifacts/benchmark_evidence_v1/reconciliation.json analysis/build_benchmark_evidence.py
37	methods-marker-strength-matching	To quantify where human-curated marker genes fall in the differential expression ranking, we matched each marker to its corresponding entry in the dataset's DEG list. For human-curated markers, matching was performed on the four-part key (dataset, data_id, group_name, feature_id), requiring exact agreement on the cell type, gene, and supplementary data source. Markers that could not be matched—because the gene was absent from the DEG list or lacked an Ensembl ID—were excluded from this analysis.	analysis/gen_strength.ipynb
38	methods-marker-rank-summary	For each matched marker, we recorded its rank within the cell type's DEG list (rank 1 = highest log-fold change) and its log-fold change. We computed the fraction of markers falling within the top N DEGs at every threshold N from 1 to 200, reported the values at $N = 50$ and $N = 100$, and separately tabulated recovery for text-sourced and image-sourced markers.	analysis/lfc_comparison.ipynb
39	methods-optimal-deg-depth	To estimate the optimal DEG list depth for recovering each marker set, we computed precision–recall curves by varying a rank threshold N from 1 to 20,000. At each threshold, we defined the top N DEGs as predicted markers and computed precision, recall, and the F-score. The optimal N was defined as the threshold that maximized the F-score. This analysis was performed separately for each dataset and marker source (text or image).	analysis/artifacts/fig2_optimal_deg_cutoff.tsv analysis/build_fig2_marker_recovery.py
40	methods-llm-run-settings	We evaluated three approaches to LLM-based marker gene curation across the seven benchmark datasets. All experiments used Claude (claude-opus-4-6) with temperature 0 and a maximum output length of 32,000 tokens; we did not benchmark model-to-model variation. Costs were computed from the model's API pricing (\$5 per million input tokens, \$25 per million output tokens for claude-opus-4-6).	analysis/build_fig2_marker_recovery.py analysis/selection_analysis.ipynb
41	methods-llm-generation	In the generation experiment (celltypes-to-genes), we gave the model the list of cell type names from each dataset's human-curated marker set and asked it to generate 5–15 well-established marker genes per cell type. The model received only the species and cell type names—no manuscript text, figures, or supplementary data. In a second generation experiment (genes-to-celltypes), we extracted the gene list for each cell type from the human-curated set, replaced the cell type names with anonymous labels (Group 1, Group 2, ...), and asked the model to predict the cell type for each group. We measured prediction accuracy as the fraction of exact string matches between predicted and original cell type names. Non-exact predictions were counted as related when they showed substring containment or shared label tokens; remaining predictions were counted as having no lexical relation.	analysis/ext_vs_gen.ipynb
42	methods-llm-selection	In the selection experiment, we parsed each dataset's DEG tables, ranked genes per cell type by absolute log fold change, and presented the top N genes (with log fold change and adjusted p -value) to the model, asking it to select which are true markers. We swept $N \in \{10, 20, 30, 40, 50, 100, 200, 300, 400, 500\}$ for each dataset and reported the N that maximized pair-level F1. As a control for whether cell type identity aids selection, we repeated the full sweep with cell type names replaced by anonymous cluster labels (Cluster 1, Cluster 2, ...), preserving the gene lists and statistics but removing cell type identity from the names.	analysis/selection_analysis.ipynb analysis/build_fig2_marker_recovery.py

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#	ID	Claim	Sources
43	methods-llm-extraction	In the joint extraction experiment, Claude received the full manuscript and a catalog mapping each candidate <code>data_id</code> to the cell type labels present in that paper's DEG sources. For every reported human marker association, the model returned the cell type, gene, verbatim supporting text, and one candidate <code>data_id</code> , or null when no source was supported. The model did not receive the human annotations. When it returned null, an exact target-label match supplied the narrowest candidate source; this affected five of 152 retained claims. We converted the output to the current <code>mrkr</code> claim structure and coalesced records with the same source text and target into marker panels.	analysis/evaluate_mrkr_joint_deg_extraction.py analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv
44	methods-llm-evaluation-subset	The extraction evaluation used positive human text annotations. Validation required every retained claim to have an exact manuscript offset and every retained marker gene to occur explicitly in that source text. Unsupported genes and claims with ambiguous source offsets were excluded. Gene labels were mapped programmatically with the packaged <code>gmap.txt</code> . We computed macro-averaged precision, recall, and F1 across studies for exact normalized cell type-gene pairs and exact <code>data_id</code> -cell type-gene triples. The same extracted terms were used for both scores.	analysis/evaluate_mrkr_joint_deg_extraction.py analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_summary.json analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_pair_review.tsv analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_triple_review.tsv
45	ext-results	Validation retained 152 associations listing 311 mapped marker genes and excluded five as unsupported or ambiguous.	analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_papers.tsv analysis/evaluate_mrkr_joint_deg_extraction.py
46	methods-llm-source-verification	The joint experiment saved each raw model response before validation. Every derived artifact records hashes for the prompt, manuscript, human benchmark, DEG catalog, and gene map. The run is therefore resumable and fails if an input changes.	analysis/evaluate_mrkr_joint_deg_extraction.py analysis/artifacts/mrkr_benchmark_pilot_20260721/joint_deg_extraction_v1/joint_extraction_report.md
47	methods-biorxiv-mirror	We selected bioRxiv preprints from a local mirror of 428,741 MECA archives downloaded from the bioRxiv source repository in January 2025. Each archive contains JATS XML and structured metadata.	analysis/biorxiv_summary.ipynb analysis/build_marker_corpus.py
48	methods-biorxiv-filter	These filters produced 504 bioRxiv manuscripts.	analysis/biorxiv_summary.ipynb analysis/build_marker_corpus.py
49	methods-hca-publication-corpus	Of 515 linked publication records, 475 had manuscript text that could be normalized to Markdown and processed.	analysis/compile_hca_human_publications.py analysis/download_hca_manuscripts.py analysis/unify_hca_manuscripts.py data/hca/manuscripts_manifest.tsv
50	methods-corpus-mrkr-workflow	For each paper, <code>mrkr extract</code> returned exact source text, normalized labels, and term offsets. <code>mrkr ground</code> mapped human genes with the packaged gene map, organisms to NCBI Taxonomy, cell types and comparisons to Cell Ontology, and tissues to UBERON. <code>mrkr validate</code> then checked the source hash, source offsets, explicit term offsets, required term counts, gene direction, and grounding metadata.	analysis/mine_corpus.py analysis/mrkr.lock analysis/artifacts/mrkr_corpus_onto_v2.tar.gz.sha256
51	methods-corpus-repair	We repaired these files without another model call. The repair removed only the failing association or conflicting gene term, recorded every removal, and reran grounding and validation.	analysis/repair_corpus.py README.md

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#	ID	Claim	Sources
52	methods-matched-label-controls	For each pair of papers that used the same cell type label, we sampled one pair with different labels. Control panels matched the original paper source and the base-2 marker-count bin. We required different article keys and repeated the control sampling with 25 fixed random seeds.	analysis/analyze_mrkr_corpus.py analysis/artifacts/mrkr_corpus_analysis_v1/matched_background_seed_sensitivity.tsv
53	methods-ontology-matched-analysis	For every such pair, we sampled three references matched by paper source and the unordered pair of base-2 marker-count bins: a same-label pair with the same accepted identifier, a same-label pair without an accepted identifier, and a pair with different identifiers. Because grounding is deterministic from the normalized label, no same-label pair had two different accepted identifiers. We gave each identifier-recovered pair equal weight. Intervals are the 2.5th and 97.5th percentiles from 1,000 pair-bootstrap and matched-sampling replicates. An ontology-term-balanced analysis is retained as a sensitivity analysis.	analysis/analyze_mrkr_corpus.py analysis/artifacts/mrkr_corpus_analysis_v1/identifier_recovered_matched_bootstrap.tsv analysis/artifacts/mrkr_corpus_analysis_v1/stable_identifier_linkage_summary.tsv
54	methods-figure-one-examples	The matrices in Supplementary Figure S2 are schematic. The myeloid examples are validated associations from the corpus, selected to illustrate relations between reported labels and marker gene panels. No corpus estimate is computed from these examples.	docs/paper/biorxiv/tex-figures/fig_paper_celltype_joint.tex docs/paper/biorxiv/tex-figures/fig_paper_celltype_joint_body.tex
55	methods-cap-liver-expression-analysis	We used CAP dataset 1440 as the liver myeloid analysis and CAP dataset 1437 as the broader all-cell liver analysis. The files are nested, not independent. All 40,821 cell barcodes in dataset 1440 occur in the 167,598-cell all-cell file, which contains 17 author cell type labels.	analysis/artifacts/cap_liver_nesting_check.tsv analysis/verify_cap_nesting.py
56	methods-cap-label-agreement	The two annotations disagree for some cells: 83.8% of the myeloid cells carry their mapped broad label in the all-cell analysis. As a control, we recomputed the all-cell one-vs-rest rankings with each target group defined by the exact barcodes of the corresponding myeloid label instead of the study's all-cell labels; mean top-100 recovery was 21.7%, matching the label-based 21%.	analysis/artifacts/cap_liver_nesting_check.tsv analysis/artifacts/cap_liver_barcode_constant_recovery.tsv analysis/build_cap_liver_barcode_constant_deg.py
57	methods-formalization-setup	We encoded the formal results in Lean 4 under <code>analysis/formal</code> . The binary model records studies, partitions, cell groups, pairwise comparisons, marker genes, and marker gene panels. The score model also records real-valued marker scores, thresholds, weighted backgrounds, and per-group mean summaries.	analysis/formal/MarkerIdentifiability/Basic.lean analysis/formal/CLAIMS.md analysis/formal/lakefile.lean analysis/formal/lean-toolchain docs/paper/biorxiv/build/supplementary-note.lean
58	methods-formalization-theorems	The Lean development proves the marker-family intersection theorem, monotonicity under restriction of the comparison groups, local-to-global and hierarchy results, equivalence between the all-pair conjunction and the binary marker-family definition, marker panels as hitting sets, panel monotonicity, weighted-background identities, score-thresholded results, and examples in which a cell type-gene projection loses comparison information.	analysis/formal/MarkerIdentifiability/Basic.lean analysis/formal/CLAIMS.md docs/paper/biorxiv/supplementary-note.tex

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